

MIX GRADE	DESIGN REF. CODE	REVISION	STATUS
M20	OR20F207	00	APPROVED
MIX DESCRIPTION	CREATED	APPROVED	
M 20 (OPC + FLY ASH) 320Kg	25 Aug 2026 Pallavi Nandakumar	25 Aug 2026 Quality Control Engineer	

MIX DESIGN SUMMARY		
Compressive Strength (f'ck) at 28 days 20.0 MPa	Standard Deviation (SD) 4.0 MPa	Maximum Aggregate Size 20 mm
Target Mean Strength (f'ck + 1.65 x SD) 26.6 MPa	Free Water / Binder Ratio (W/B) 0.469	Water Content 150 kg/m3
Total Binder (Cement + Fly Ash) 320 kg/m3	Target Workability at Site 100+/-25mm	Design Density 2,466 kg/m3

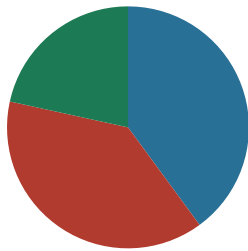
MIX PROPORTIONS (SSD/m3)	
Cement	250 kg/m3
Fly Ash (PFA) — 22% of Binder	70 kg/m3
Total Binder (Cement + Fly Ash)	320 kg/m3
Free Water Content	150 kg/m3
Free Water / Binder Ratio (W/B)	0.469
Fine Aggregate	796 kg/m3
Coarse Aggregate — 20 mm	768 kg/m3
Coarse Aggregate — 12.5 mm	430 kg/m3
Total Aggregate	1,994 kg/m3
Admixture — Asian PC-200, 0.59% of Binder	1.90 kg/m3
Design Density	2,466 kg/m3

MATERIALS — TYPE / SOURCE / SP. GR.		
MATERIAL	TYPE / SOURCE	SP.GR.
Cement	OPC 53 Grade / Dalmia / Ramco	3.15
Fly Ash (PFA)	Fly Ash (PFA)	2.20
Fine Aggregate	Manufactured Sand (M-Sand), Zone II / III	2.75
Coarse Aggregate — 20 mm	Crushed	2.72
Coarse Aggregate — 12.5 mm	Crushed	2.71
Admixture	Asian PC-200	1.10
Maximum Aggregate Size	20 mm	—

AGGREGATE PROPORTIONS		
FRACTION	QTY (kg/m3)	% OF TOTAL
Fine Aggregate	796	40%
Coarse — 20 mm	768	39%
Coarse — 12.5 mm	430	22%
Total	1,994	100%

Coarse / Fine Aggregate Ratio	
COARSE 60%	FINE 40%

Aggregate Proportion



- M-Sand (fine) — 40%
- 20 mm (coarse) — 39%
- 12.5 mm (coarse) — 22%

NOTE
1. Admixture dosage to be adjusted based on ambient temperature and transport time. 2. Moisture correction of aggregates to be carried out daily. 3. All quantities are in SSD condition unless specified.

Quality Control Engineer
Quality Control, Our Own Ready Mix

Disclaimer: This mix design has been prepared in accordance with applicable Indian Standards based on laboratory trials and the materials specified herein. Actual field performance is dependent upon proper batching, transportation, placement, compaction, finishing, curing and site conditions. Any change in material source or characteristics may require review and revision of the mix design.